

REVIEW

Nicotine-Mediated Redox and Nitrosative Stress in Cardiorenal Health

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ABSTRACT: Nicotine exposure is increasingly widespread due to the expansion of the use of noncombustible tobacco and nicotine-replacement products; yet nicotine is still being framed as an addictive but relatively benign compound. This review synthesizes clinical and experimental evidence linking nicotine exposure to cardiorenal disease, emphasizing that risks persist after cessation. We discuss here a central novel concept that damage caused by nicotine in tissues is not determined solely by receptor signaling, as nicotine exists in chemical forms capable of strong receptor-independent membrane diffusion. Mechanistically, we highlight convergent pathways of nicotine-induced cellular injury, focusing on redox stress and underscoring major limitations in the field, such as the lack of studies in epithelial cells. We further emphasize the emerging paradigm that links nicotine and nitric oxide synthase (NOS) biology, in which reduced nitric oxide bioavailability, NOS uncoupling, and peroxynitrite-mediated protein nitration represent underappreciated drivers of persistent cardiorenal injury in response to nicotine. Finally, we outline gaps and research priorities needed to identify therapeutic strategies that specifically mitigate nicotine-driven oxidative and nitrosative injury in vulnerable populations with hypertension, diabetes, and chronic kidney disease.

Key Words: hypertension ■ kidney ■ nicotine ■ nicotine replacement therapy ■ nitric oxide ■ oxidative stress

Despite the fact that nicotine is marketed as an addictive but less harmful than smoking compound, clinical and experimental evidence points to nicotine as a key driver of cardiovascular and renal damage. Smokers are more likely to develop hypertension and adverse cardiovascular events, including stroke and heart failure. In 2024, the American Heart Association issued an updated statement¹ on the use of smokeless tobacco products, raising additional concerns regarding oral nicotine products, such as gums, lozenges, and tablets. Cross-sectional analysis has established that use of smokeless products, such as vapes, causes an increase in blood pressure between 5 and 10 mmHg and an overall increase in cardiovascular risk^{1,2} similar to harm from traditional cigarettes. Smokers exhibit higher rates of albuminuria and faster chronic kidney disease (CKD) progression³ (Figure 1). Effects of nicotine products consumed orally on cardiovascular and renal health are not established.¹

In January 2025, the US Food and Drug Administration (FDA) proposed the development of a tobacco product standard that would regulate nicotine yield by

establishing a maximum nicotine level in cigarettes and certain other combusted tobacco products (FDA-2024-N-5471). Interestingly, nicotine, the main chemical in tobacco and many smoking cessation products, has been presented as an addictive but benign compound for decades.⁴ This shifts basic and clinical research focus away from direct pathological effects of nicotine, validating the development of alternative ways of nicotine consumption (vaping, nicotine pouches, heating tobacco, chewing tobacco) while ignoring the harmful properties of nicotine itself. The main sources of nicotine exposure could be presented as follows:

- Combustible tobacco products: cigarettes, cigars, cigarillos, hookahs, and pipe tobacco.
- Noncombustible tobacco products: e-cigarettes, heat-not-burn products that use real tobacco but produce an inhalable aerosol instead of smoke, smokeless tobacco like nicotine pouches, chewing tobacco, snus, and dip.
- Nicotine-replacement products (NRPs): nicotine gums, patches, lozenges, inhalers, and nasal sprays.
- Passive nicotine exposure.

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Nonstandard Abbreviations and Acronyms

BH₄	tetrahydrobiopterin
CKD	chronic kidney disease
CORAL	Cardiovascular Outcomes in Renal Atherosclerotic Lesions
FDA	US Food and Drug Administration
GTPCH1	GTP cyclohydrolase-1
nAChR	nicotinic acetylcholine receptor
NO	nitric oxide
NRP	nicotine-replacement product
OCT	organic cation transporter
ROS	reactive oxygen species
SRNG	smoking-related nodular glomerulopathy

According to the 2022 National Health Interview Survey data, 19.8% (1 in 5) of US adults reported current tobacco product use. According to this data, cigarettes remain the most commonly used type of tobacco product, and e-cigarettes are next, with an increased use among adults since 2019. Vaping, presented as a healthier alternative to cigarettes, has become popular among the younger populations. Across age groups in the United States, adults aged 18 to 24 years have the highest

prevalence of e-cigarette use, and adults aged 45 to 64 years have the highest prevalence of cigarette smoking.⁵ Moreover, in 2024, 19.0% of US middle and high school students (representing 5.28 million students) reported ever having used any tobacco product, and 8.1% (2.25 million students) reported current use of any tobacco product,⁶ with the majority of them using e-cigarettes. Disposable vaping devices, which are popular among students, often have high nicotine concentrations; moreover, the manufacturers of these devices are undefined, with no regulation standards available. For example, one of the most popular brands of disposable vaping devices in the United States, contains 5% nicotine,⁷ and according to the 2024 National Youth Tobacco Survey, 36.1% of youth use these particular vapes. In 2018, 20.9% of US adults were reported to be former smokers, and the National Health Interview Survey reports that the ratio of former to current smokers in the United States is 3:2. While comprehensive national statistics on nicotine exposure are not available, these data allow to suggest that almost half of the US population is currently or was in the past exposed to nicotine, while the long-term mechanistic effects of nicotine exposure are severely understudied.

While combustible and most noncombustible tobacco products are FDA-controlled, NRPs are FDA-approved,

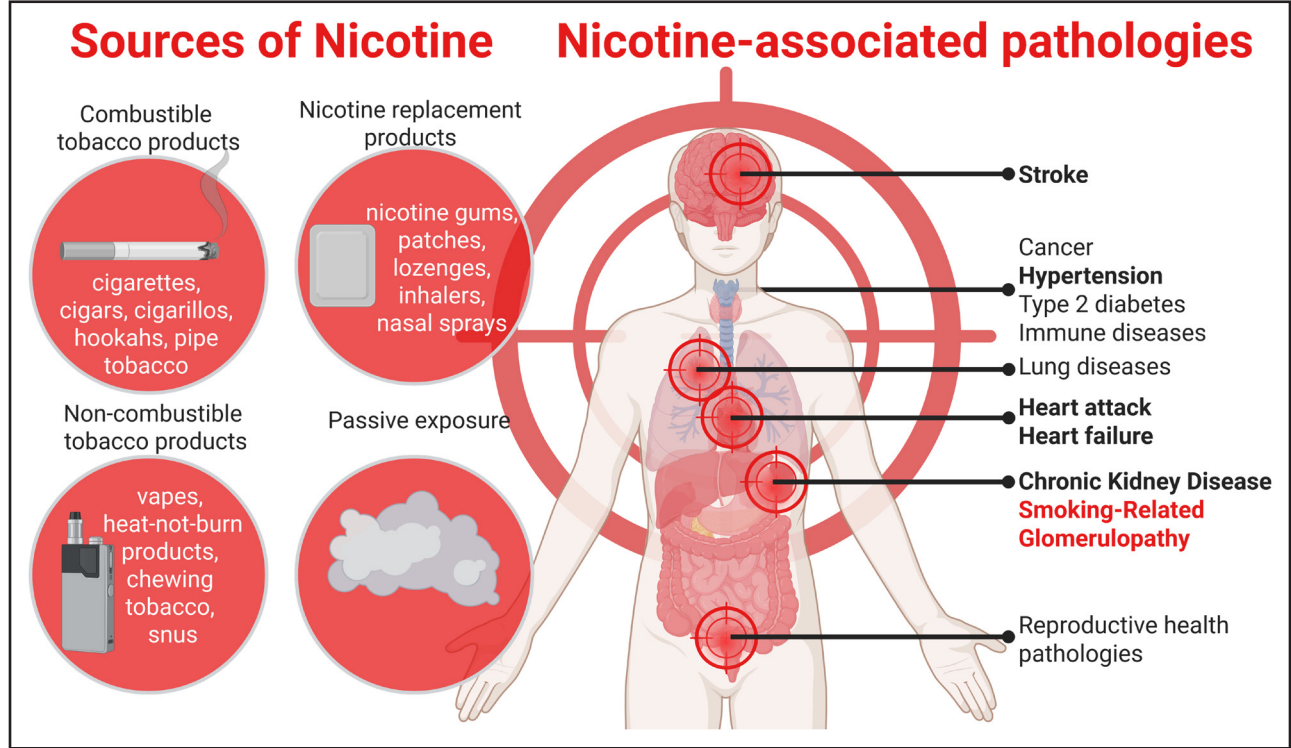


Figure 1. Sources of nicotine exposure and associated renal and cardiovascular pathologies. Nicotine exposure occurs through multiple sources, including combustible tobacco products, noncombustible products, such as e-cigarettes/vapes, nicotine-containing replacement therapies, and through passive exposure. Nicotine exerts widespread systemic effects, contributing to cardiovascular and cerebrovascular disease (hypertension, heart attack, heart failure, and stroke), metabolic and immune disorders (type 2 diabetes and immune dysfunction), pulmonary disease, cancer, reproductive health pathologies, and chronic kidney disease, including smoking-related nodular glomerulopathy. Created in BioRender. Palygin, O. (2026) <https://BioRender.com/lsmamq0>.

which further supports the public perception of them as a safe alternative to smoking. The perception of nicotine as a harmless substance has led to NRPs being extensively used. Interestingly, the 5-year success rate of smoking cessation programs is reported to be only 20% to 30%.⁸ A more recent study did not find consistent evidence that NRP use is related to the long-term success of smoking cessation; most importantly, the study showed that smokers remain interested in using NRPs even 5 years after the completion of the smoking cessation program.⁹ The Table highlights the substantial amounts of daily nicotine exposure across smoking products, including NRP nicotine-containing substances. Conservative estimations suggest that conventional cigarettes deliver the highest nicotine burden, with average smokers reaching $\approx$ 1.7 to 2.4 mg/kg per day and heavy smokers exceeding 4 to 7 mg/kg per day, reflecting both high nicotine content per cigarette and frequent daily use.¹⁰ Vaping results in lower but still significant exposure, particularly in heavy users, where nicotine intake can surpass 1.5 mg/kg per day depending on e-liquid concentration and puff frequency.¹¹ Among NRPs, nicotine patches provide the lowest daily doses (generally $<$ 0.3 mg/kg per day), while gum and lozenges yield higher exposure (up to 40–80 mg/kg per day) but remain below that of cigarettes and heavy vaping. Overall, while combustible tobacco products produce the highest systemic nicotine exposure, and vaping occupies an intermediate range with high variability, Centers for Disease Control and Prevention (CDC)-recommended NRP regimens still deliver relatively high and persistent nicotine doses. Thus, one could surmise that NRPs are generally associated with lower consumption if taken according to the manufacturer's instructions; unfortunately, on the contrary, the use of NRPs often leads to continuous nicotine consumption without a clinical smoking risk status typically applied to smoking and vaping patients.¹² Moreover, in nephrology, exposure to NRPs in kidney donors and transplant patients generally does not qualify as smoking

in clinical charts, is not being taken into consideration when transplant quality and patient suitability are being evaluated, and thus poses a hidden risk. While we can usually estimate patients' nicotine exposure when they use combustible and noncombustible tobacco products, it is very hard to assess NRP and passive nicotine exposure,¹³ since these sources of nicotine are not perceived as dangerous as smoking and often are not reported to the health provider.

WHY SMOKING CESSATION ALONE FALLS SHORT: A NEPHROLOGIST'S VIEW

Despite decades of cessation encouragement efforts, in 2021 in the United States, 22.0% of people aged 12 years or older reported nicotine consumption via tobacco products or vaping (2021 National Survey on Drug Use and Health). While quitting is a critical step, damage elicited by nicotine persists for decades after cessation.^{14,15} Rodent studies have linked nicotine to CKD,^{16–18} and prospective and population-based studies have confirmed that nicotine perpetuates kidney pathology and augments the risk of proteinuria.^{19–21} Prospective studies have also shown that secondhand smoke exposure increases the risk of CKD.^{22,23} In subjects with hypertension and patients with preexisting renal disease, smoking-related kidney pathology is even more striking.^{24–29} Although smoking cessation remains the cornerstone of risk reduction, growing evidence indicates that cessation of nicotine exposure alone is often insufficient to reverse established end-organ damage, particularly in the cardiovascular and renal systems. This finding is possibly connected to the limited ability of renal cells to regenerate and recover after chronic injury (see more detailed discussion below in the Smoking-Related Nodular Glomerulopathy section). Epidemiological and experimental studies demonstrate that prior smoking is associated with persistent cardiovascular risk, endothelial dysfunction, and accelerated CKD progression long after

Table. Smoking Product Consumption and Variable Sources of Nicotine

Exposure type	Conventional smoking and vaping		Over-the-counter nicotine-replacement therapy		
	Cigarette	Vaping	Nicotine patch	Nicotine gum	Nicotine lozenge
Nicotine content in tobacco products	8–20 mg (avg: 12)	3–60 mg/mL (avg: 30)	7, 14, or 21 mg	2.0 or 4.0 mg	2.0 or 4.0 mg
Average amount consumed per day (average consumer)	10–14 cigarettes (CDC, 01/2018)	100–200 puffs ($\approx$ 1–2 mL)	1 $\times$ 24-h 7 or 14 mg patch	8–12 pieces of 2.0 mg or 4–6 pieces of 4.0 mg	1 $\times$ 2.0 mg every 1–2 h or 5 every 6 h, 9 to 20 per day
Average amount consumed per day (heavy consumer)	$>$ 25 cigarettes ¹⁰	365 puffs ¹¹	1 $\times$ 24-h 14 mg or 21 mg patch	13–20 pieces of 2.0 mg or 7–10 pieces of 4.0 mg	1 $\times$ 4.0 mg every 1–2 h or 5 every 6 h, 9 to 20 per day
Nicotine exposure range (average consumer), mg/kg per d	1.7–2.4	0.4–0.9	0.1–0.2	0.2–0.3	0.3–0.6
Nicotine exposure range (heavy consumer), mg/kg per d	4.2–7.14	$>$ 1.5	0.2–0.3	0.4–0.6	0.5–1.1

Average nicotine exposure ranges calculated for a 70-kg person. Daily nicotine consumption derived from average (12 mg per cigarette or 30 mg/mL of e-liquid) and heavy (12–20 mg per cigarette or 30–60 mg/mL of e-liquid) consumers. Nicotine-replacement therapy values are assembled from Centers for Disease Control and Prevention (CDC) approved treatment recommendations, and do not include calculations for individuals that augment or deviate from prescribed treatment strategies.

cessation, consistent with durable structural, metabolic, and epigenetic remodeling rather than purely acute toxic effects.^{30–33} Importantly, there are no approved therapies that specifically target nicotine-induced cardiorenal injury, and individuals attempting to quit continue to be exposed to pharmacological nicotine through NRPs, which, while reducing combustion-related toxins, may perpetuate nicotine-driven cellular stress pathways. This issue is particularly relevant for patients with hypertension, diabetes, and established cardiovascular or kidney disease, who exhibit heightened susceptibility to nicotine-induced oxidative, inflammatory, and metabolic injury and experience worse outcomes compared with nonsmokers.^{34,35} This emphasizes a critical gap in our understanding of nicotine-mediated damage and highlights the need to define new disease-modifying pathways tailored to nicotine-exposed populations, where nicotine-associated pathology may be biologically distinct from that observed in otherwise healthy smokers.

NICOTINE: BEYOND RECEPTOR BINDING

Textbook biological actions of nicotine are exerted by binding to nicotinic acetylcholine receptors (nAChRs), which are pentameric ligand-gated cation channels.³⁶ Nicotine binds to an extracellular orthosteric site; receptor activation opens a pore that conducts Na^+/K^+ and often Ca^{2+} , driving depolarization and Ca^{2+} -dependent signaling. Importantly, nAChRs are signaling receptors; they do not function as nicotine transporters. Nicotine or acetylcholine are ligands that bind and gate the channel, and transported Ca^{2+} then further acts as a potent intracellular signal transducer. Both vascular and renal human cells express nAChRs, but their composition and function are still under investigation. There are 16 nAChR subunits identified, including 10 α , 4 β , as well as γ , δ , and ϵ nAChR subunits, that form pentameric nAChRs in various combinations.³⁷ The Staruschenko Lab Hypertension Atlas,³⁸ which reports RNA-seq data from rat kidney, heart, and liver, detected expression of nAChRs in renal and heart tissues. Both renal cortex and medulla exhibited the highest abundance of *Chrna2*, *Chrna7*, *Chrnb1*, *Chrnb2*, and *Chrne* genes, encoding nAChR α 2, nAChR α 7, nAChR β 1, nAChR β 2, and nAChR ϵ subunits; and heart tissue analysis presented the highest abundance of *Chrna7*, *Chrna4*, *Chrnb1*, *Chrnb2*, and *Chrng*, encoding nAChR α 7, nAChR α 4, nAChR β 1, nAChR δ , and nAChR γ subunits. The different subunit distribution could result in ionotropic or metabotropic responses to nicotine or acetylcholine and variation in the action of nAChRs in different tissues. However, this conventional knowledge needs to be reviewed in the context of more intricate and often overlooked information regarding receptor-independent routes of nicotine entry into the cells.³⁹ Depending on pH, nicotine can exist in 3 interconvertible forms: freebase (unprotonated), monoprotonated, and diprotonated

nicotine⁴⁰ (Figure 2). Diprotonated nicotine is prevalent under highly acidic conditions. This form is low in most tobacco or vaping products because such a low pH is rarely reachable. In practice, only freebase and monoprotonated nicotine contribute to nicotine absorption in the human body. Of note, freebase nicotine is uncharged and lipophilic, readily crossing the cell membrane in seconds.⁴¹ In contrast, protonated forms are charged and more water-soluble, crossing membranes more slowly and less efficiently (Figure 2). In tobacco leaves, nicotine is mostly present in the protonated form, which is not as readily bioavailable. However, combustible tobacco products have long been engineered to create alkaline pH forms (with extreme alkalinity up to pH 11 in vaping products) and optimize nicotine to improve cell absorption. In cigarettes, the fraction of freebase nicotine in cigarette smoke varies by brand, ranging from a few percent up to 50%.⁴² Smokeless tobacco and NRPs (eg, chewing tobacco and nicotine gums) are buffered to an alkaline pH to increase the freebase concentration and facilitate nicotine absorption through mucous membranes and the skin.⁴³ Vapes are the highest-pH tobacco products, and thus a higher freebase nicotine concentration has been associated with greater nicotine absorption and greater cardiovascular effects.⁴⁴ Vaping products allow the delivery of mainly freebase nicotine (up to 95%), with nicotine salts (monoprotonated form) being incorporated as a smoother alternative, as freebase nicotine decreases inhalability.⁴⁵ Thus, the proportion of nicotine in freebase form compared with protonated form, as well as patient acid-base status, can influence how quickly and how much nicotine is absorbed during smoking or vaping. Alternatively, at physiological pH, when a substantial fraction of nicotine is protonated, an organic cation transporter (OCT) system may become key for nicotine transport in specific tissues (notably the kidney and liver). Particularly, the Solute Carrier Family 22 (SLC22) OCT family (for instance, OCT2/SLC22A2 in the kidney) is proposed as a determinant of renal uptake of nicotine: human genetic variation in OCT2 measurably impacts nicotine/cotinine pharmacokinetics in controlled infusion studies, supporting the contribution of this transporter to nicotine handling.^{46,47} According to Susztak Lab Kidney Biobank data,⁴⁸ all renal cells, including podocytes, express these transporters, with the highest abundance found in tubular cells.

Collectively, these observations demonstrate that cellular nicotine exposure is not dictated solely by nAChR engagement, but by a complex interplay between nicotine chemistry, product formulation, acid-base balance, tissue pH, and transporter abundance. Considering the observations outlined above, individuals with metabolic or respiratory alkalosis and salt-sensitive (low-renin) hypertension who consume vaping products constitute a highly vulnerable, high-risk population. Receptor-independent membrane diffusion and OCT-mediated

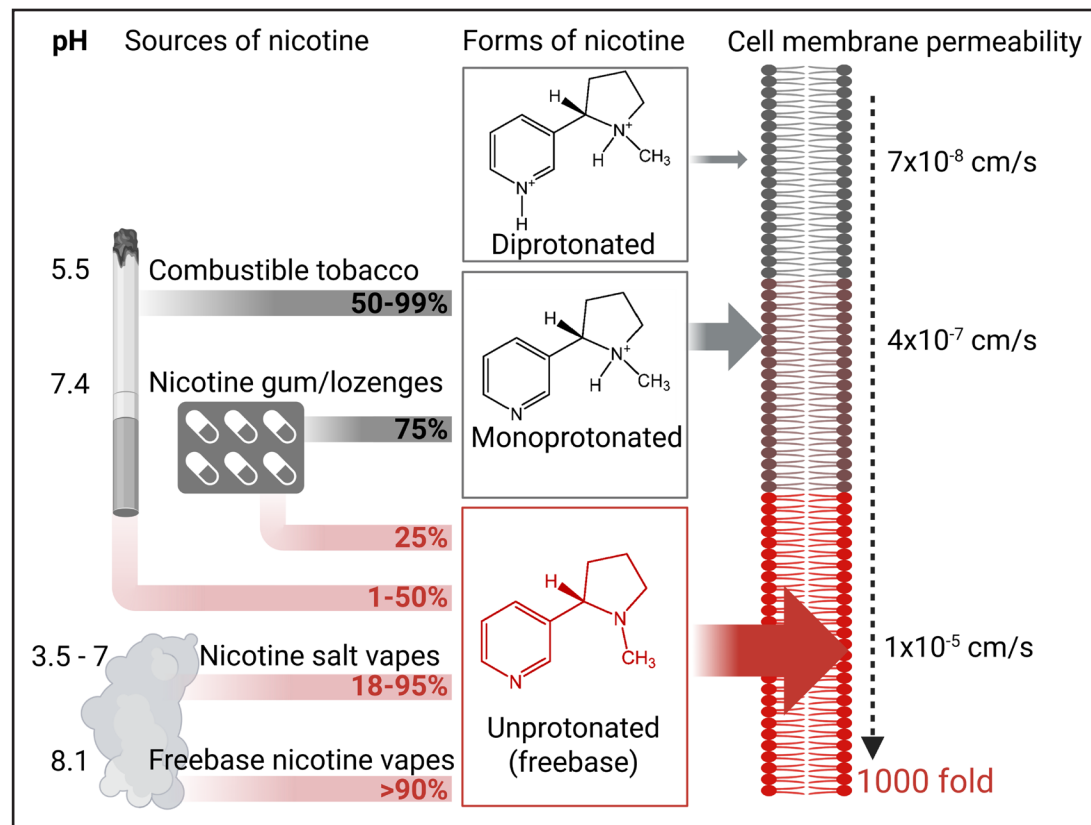


Figure 2. Source-dependent nicotine form determines membrane permeability.

Nicotine exists in diprotonated, monoprotated, and nonionized (freebase) forms, with the relative abundance of each species determined by pH. Combustible tobacco products predominantly deliver monoprotated nicotine, which exhibits slower passive diffusion across lipid bilayers. In contrast, nicotine salts and freebase nicotine used in e-cigarette formulations predominantly contain a nonionized form of nicotine, resulting in a markedly higher fraction (>90%) of membrane-permeable nicotine.⁴¹ Importantly, nicotine-replacement therapy products, such as nicotine gums and lozenges, contain a significant fraction (25%) of unprotonated nicotine. Created in BioRender. Palygin, O. (2026) <https://BioRender.com/tspc2gw>.

uptake can substantially shape intracellular nicotine burden, particularly in highly perfused and excretory organs such as the kidney. Appreciating these noncanonical entry routes is essential for understanding nicotine's systemic and tissue-specific effects, including redox toxicity that cannot be explained by receptor signaling alone.

NICOTINE'S EFFECTS ON CARDIOVASCULAR AND RENAL HEALTH

Autopsy case reports suggest that major targets for nicotine exposure and accumulation in the human body are blood vessels, the brain, and the kidney.⁴⁹ This correlates with clinical findings that for smoking patients, the time of the first stroke onset is 10 years earlier than for non-smoking patients.⁵⁰ Moreover, the kidneys are responsible for nicotine clearance.⁵¹ Less than 5% of circulating nicotine is protein-bound,⁵² thus nicotine is easily transferred through the glomerular filtration barrier, leading to wide exposure of the renal cells to nicotine. Approximately 10% of absorbed nicotine is excreted in urine as is, with the remainder appearing as metabolites (primarily

cotinine and hydroxylated products).^{53,54} In rodent models, nicotine exposure leads to measurable cotinine in kidney tissue at levels similar to plasma concentrations seen in smokers (eg, ≈ 60 – 130 ng/mL plasma cotinine); kidneys show biologic responses (oxidative stress and pro-fibrotic signaling) consistent with local tissue exposure, which implies that nicotine itself or its metabolites accumulate sufficiently in renal tissue to activate local pathways.⁵⁵ In addition, a cross-sectional data analysis revealed a negative correlation between nicotine metabolism rate and kidney function,⁵⁶ suggesting that exposure to nicotine is detrimental to kidney cells. Of note, the mechanisms of this damage are not well understood.

Nicotine consumption has adverse effects on overall cardiovascular health. In 2021, the study by Khan et al, using data from 106 165 adults in 9 population-based cohorts in the United States, reported long-term risks for cardiovascular events as 46.0% and 34.7% in middle-aged men and women, respectively, with higher competing hazard ratios for the first presentation being a fatal event in smokers than in nonsmokers.⁵⁷ It is well established that nicotine impacts the cardiovascular system by promoting sympathetic overactivation, oxidative stress,

and endothelial dysfunction. Smoking was strongly associated with worse cardiovascular and kidney disease outcomes in several observational studies.^{28,58} Its effects collectively contribute to multiple cardiac and renal pathologies, including vascular (ie, atherosclerosis and stroke), myocardial issues, hypertension and heart failure, arrhythmias, CKD, and increased susceptibility to cardiovascular diseases overall.⁵⁹ Data from the Study of Heart and Renal Protection show that smoking significantly increased the risks for vascular and nonvascular morbidity and mortality, although it was not associated with end-stage kidney disease development.³ However, it should be noted that current smokers enrolled in the Study of Heart and Renal Protection had relatively low cigarette consumption (average, 10 cigarettes per day), and smoking status was only assessed at the starting point. The Multiple Risk Factor Intervention Trial, which included heavy smokers (>30 cigarettes per day) reported an association between smoking and end-stage kidney disease.⁶⁰ Moreover, population studies have also shown an increased risk of end-stage kidney disease development in current smokers compared with patients who never smoked.⁶¹

It is established that nicotine induces an elevation of blood pressure and heart rate, as well as dysregulation of cardiac function, by activation of the sympathetic nervous system with the release of catecholamines such as norepinephrine.⁶² Norepinephrine binds to $\alpha 1$ and $\alpha 2$ adrenergic receptors in vascular smooth muscle, facilitating Ca^{2+} ion influx and stored Ca^{2+} release, inhibiting adenyl cyclase activity, thus regulating arteriolar resistance, venous capacitance, and smooth muscle vasoconstriction.^{63,64} Nicotine in electronic cigarettes induces an increase in sympathetic nerve activity, which is accompanied by increases in blood pressure and heart rate, while decreasing baroreflex response and arterial compliance.⁶⁵ Nicotine consumption enhances myocardial contractility through nAChR-induced sympathetic activation,⁵⁹ primarily via the $\alpha 7$ -nAChR receptor, which also plays a role in blood pressure regulation through baroreceptor modulation as shown in rodent models.^{66,67} Moreover, in mice models, nicotine derived from both traditional cigarettes and electronic cigarettes induces cardiac hypertrophy in a time-dependent manner.⁶⁸ Smoking has been shown to alter both systemic and pulmonary blood pressure. Importantly, nicotine is known to damage the endothelium, reducing vasodilation and impairing blood flow: smoking contributes to endothelial dysfunction through increased oxidative stress and inflammation, which has been shown in comparative and crossover studies, as well as in human endothelial cells.^{69–71} Endothelial dysfunction contributes to impaired relaxation in both resistance and conductance vessels, which is linked to hypertension development.⁷² Prolonged administration of nicotine in animal models has been shown to precipitate damage resulting in increased blood pressure,

heart rate, cardiomyocyte inflammation, atherosclerosis, cardiac fibrosis, cardiac hypertrophy, and vascular endothelial dysfunction.^{73,74}

Nicotine has been identified as a major risk factor for kidney disease in the general population, and its effects were reported to be especially prominent in patients with diabetes and hypertension.^{75,76} The Kidney Disease Improving Global Outcomes guidelines recommend that patients with diabetes and CKD avoid tobacco products and reduce secondhand smoke exposure.⁷⁷ The Kidney Disease Improving Global Outcomes guidelines recommends smoking cessation for patients with CKD and diabetes and also highlight the sparsity of available studies evaluating the effects of tobacco products, including regular cigarettes, electronic cigarettes, and NRPs.⁷⁷ The link between smoking, progression of CKD, and a decline in renal function is well established. The Kidney Disease Improving Global Outcomes guidelines recommend offering smoking cessation programs, which include options like NRPs (which only reduce nicotine consumption), to all eligible patients with CKD and those awaiting or having received a kidney transplant.⁷⁷ Nicotine consumption and metabolism are linked to changes in eGFR, with the majority of studies showing that smoking is associated with an increase in eGFR due to compensatory hyperfiltration.^{58,78} A few studies have shown the opposite effect, which could possibly be explained by an initial decrease in renal plasma flow caused by smoking before glomeruli are damaged or by different levels of nicotine exposure.⁵⁸ Moreover, smoking is an independent factor for proteinuria in healthy subjects.^{79,80} Of note, smoking is a major risk factor for renal vascular pathology, increasing the risk of renal artery stenosis.⁸¹ Data from the Cardiovascular Outcomes in Renal Atherosclerotic Lesions (CORAL, REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT00081731) clinical trial suggest that smokers develop vascular pathology and composite end point events (defined as the first occurrence of stroke, cardiovascular or renal death, myocardial infarction, congestive heart failure, or renal replacement) at a younger age compared with nonsmokers.⁸¹

Multiple reports highlight nicotine's contribution to glomerular and tubular renal pathologies.²⁴ Documented glomerular lesions associated with nicotine consumption include smoking-related nodular glomerulopathy (SRNG, discussed in more detail below)²⁴ and nicotine-induced endothelial injury.⁸² Nicotine has also been shown to increase proteinuria, promoting albumin-induced tubular damage and oxidative stress, as well as pathological activation of $\alpha 7$ -nAChR in renal tubular cells.⁸³ The 2023 analysis of the Nephrotic Syndrome Study Network (NEPTUNE) study showed that, in patients with proteinuric glomerulopathies, the prevalence of active smoking, former smoking, and exclusive passive smoking among adults at baseline was 14.6%, 29.1%, and 4.9%, respectively.⁵⁸ Passive smoke exposure was 16.7% among

children with glomerulopathies.⁵⁸ These numbers highlight the overwhelming contribution of nicotine to glomerular pathology development.

SMOKING-RELATED NODULAR GLOMERULOPATHY

Chronic smoking in hypertensive subjects may lead to SRNG, a pathology presenting with an expanded mesangial matrix, podocyte loss, glomerular and tubular basement membrane thickening, and vascular hyalinosis^{24,84} (Figure 3). Although the association between smoking and idiopathic nodular glomerulosclerosis has been recognized for decades, only recently has smoking been identified as a driver of this distinct renal disease phenotype.⁸⁵ Traditionally, nodular glomerulosclerosis was labeled idiopathic because a clear cause was not recognized; however, recent data have highlighted smoking as its cause, shifting the perception of this disease to a smoking-associated entity. The mechanisms of SRNG are unknown; pathologically, it mimics diabetic nephropathy without clinical evidence of diabetes or alterations in glucose metabolism.^{24–29,86,87} Clinical reports have presented

SRNG as a pathological condition characterized by the onset of diabetic-like glomerulopathy in subjects with no history of diabetes.^{24–29,86} However, diffuse mesangial sclerosis can also occur in SRNG without nodules,²⁴ which is typical in diabetic nephropathy. SRNG can also present with chronic thrombotic microangiopathy, manifested through double contouring by light microscopy and subendothelial space widening by electron microscopy, as well as vascular sclerosis, medial hypertrophy, and hilar neovascularization.²⁴ SRNG was observed in seniors with a history of chronic smoking and hypertension²⁵ and in young people exposed to nicotine due to passive smoking.⁸⁶ It is strongly associated with smoking, hypertension, and obesity.²⁶ Patients with SRNG present with proteinuria and renal insufficiency, without evidence of diabetes or hyperglycemia, which serves as a key distinguishing factor between diabetic nephropathy and SRNG. Despite the similarity in glomerular pathology with diabetic nephropathy, SRNG is strongly associated with hypertension.⁸⁸

Collectively, epidemiological, clinical, and experimental studies establish strong associations between nicotine exposure and adverse cardiovascular and renal

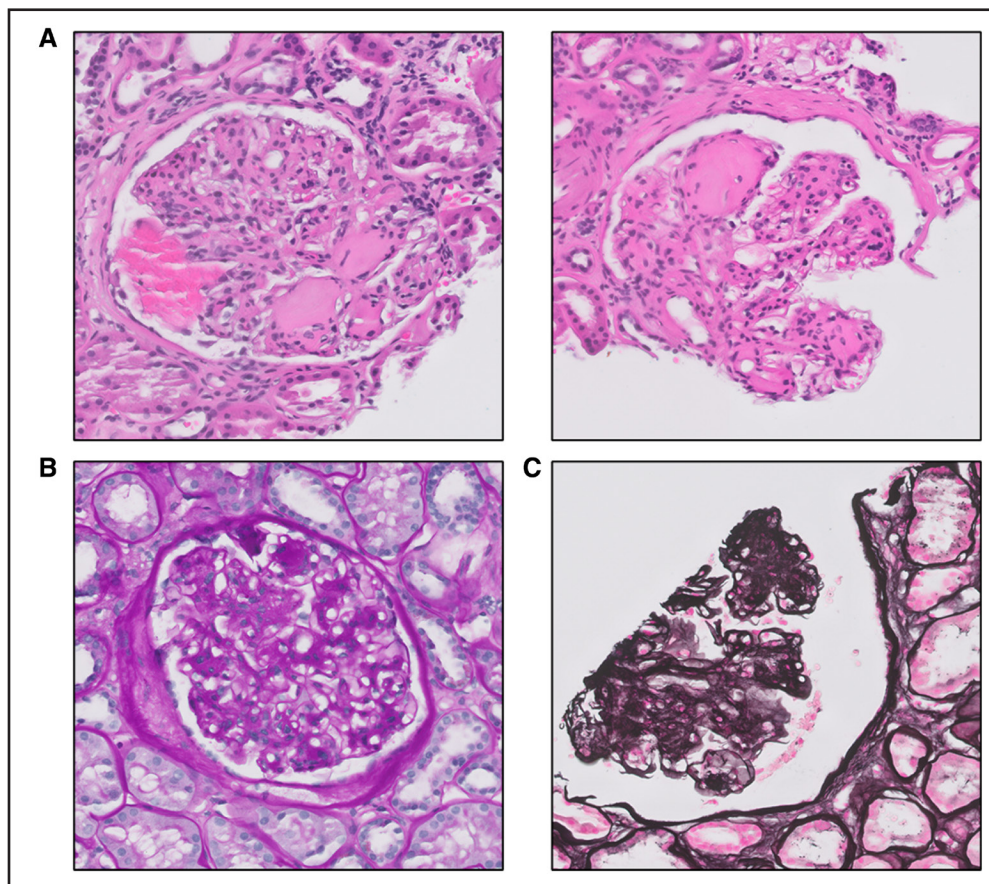


Figure 3. Idiopathic nodular glomerulosclerosis (smoking-related nodular glomerulopathy) in a female smoker.

All pictures were taken at 200× magnification. **A**, Hematoxylin and eosin staining. **B**, Periodic acid Schiff staining. **C**, Jones Silver staining. All images show prominent nodular expansion of the mesangium with mild to moderate mesangial hypercellularity. The nodular material is bright PAS-positive and argyrophilic. It is Congo red negative, and no light chain restriction was noted by immunofluorescence.

outcomes. However, despite the consistency of these observations across populations and model systems, the cellular and molecular mechanisms by which nicotine directly injures vascular, cardiac, and renal tissues remain poorly defined. This lack of mechanistic insight represents a critical knowledge gap that limits causal interpretation and constrains the development of targeted interventions to mitigate nicotine-associated cardiorenal disease.

MECHANISMS OF NICOTINE-INDUCED CARDIOVASCULAR AND RENAL DAMAGE: REDOX STRESS

Nicotine is increasingly recognized as a major determinant of oxidative stress and a disruptor of cellular metabolism; however, the literature on specific reactive oxygen species (ROS) types triggered by nicotine exposure remains limited, with most studies examining ROS species in general, rather than specific individual ROS subtypes, such as radical (superoxide anion, hydroxyl, peroxy, nitric oxide (NO) radicals, and others) versus nonradical (hydrogen peroxide) ROS. It has been hypothesized that smoking induces nodular glomerulosclerosis through the formation of advanced glycosylation end products with the induction of oxidative stress.⁸⁵ Across multiple tissues, nicotine exposure is associated with increased ROS generation, depletion of antioxidant defenses, and activation of redox-sensitive stress signaling such as mitogen-activated protein kinases (MAPKs) and nuclear factor kappa-light-chain-enhancer of activated B cells (NF- κ B) that drives inflammation, cell death, and maladaptive remodeling.^{73,89–92} Mechanistically, nicotine-linked redox stress is most consistently attributed to 2 major ROS sources, mitochondrial ROS and NADPH oxidases.^{73,89,93,94} In cultured human podocytes, terminally differentiated and nonregenerating cells, nicotine increased intracellular ROS (measured by 2',7'-Dichlorodihydrofluorescein diacetate [DCFH] dye) and triggered apoptosis through downstream stress-kinase pathways, providing a plausible direct mechanism for smoking/nicotine-associated progression of proteinuric kidney disease.^{16,91} It was further shown in diabetic nephropathy models that nicotine can amplify podocyte superoxide production and upregulate lipid-handling pathways such as CD36, linking oxidative stress to the metabolic and lipotoxic vulnerability of the glomerular filtration barrier.¹⁶ Furthermore, nicotine has been shown to promote NLRP3 inflammasome-associated podocyte injury, further connecting oxidative stress to inflammatory cell death and glomerular barrier dysfunction.⁹⁵ Chronic nicotine exposure increased renal redox stress markers in the renal cortex and worsened the ischemia-reperfusion injury phenotype in vivo, consistent with a potential redox priming effect that lowers injury thresholds.¹⁷ Interestingly, chronic nicotine administration also

augmented renal oxidative stress and damage through transcriptional activation of p66Shc (a redox-regulatory adaptor that can amplify mitochondrial ROS and apoptotic susceptibility); p66Shc was reported to potentiate nicotine-driven lipotoxic oxidative stress in renal proximal tubule cells.^{90,96} Effects of nicotine have been relatively well studied in vascular cells; it can directly alter the phenotype of human primary vascular smooth muscle cells via nAChR-linked signaling, promoting their transition from a contractile to a synthetic function, which is a remodeling program that is commonly known to be redox-sensitive.⁹⁷ Nicotine was shown to stimulate pro-calcific remodeling in human subjects, which was shown by in vitro studies to be mediated by activation of $\alpha 7$ and $\alpha 3$ nAChRs in vascular smooth muscle cells, an increase in intracellular Ca^{2+} , and NOX5-dependent oxidative stress.⁹⁴ In mice in vivo, nicotine also promoted atherosclerosis and induced endothelial pyroptosis via ROS-NLRP3 inflammasome activation in human aortic endothelial cells, further linking oxidant stress to cell death and plaque progression.⁹⁵ In a diet-induced rat model of obesity, oral nicotine administration aggravated endothelial dysfunction with increased oxidative stress and vascular inflammation, suggesting that nicotine can synergize with metabolic risk states to intensify endothelial redox injury.⁹⁸

In the heart, in vivo nicotine exposure increased myocardial mitochondrial ROS, shifted redox balance toward lower glutathione abundance, and promoted hypertrophy, fibrosis, and functional impairment; these phenotypes were attenuated by targeting mitochondrial ROS with the mitochondria-directed antioxidant mitoTEMPO, supporting mitochondrial ROS as an upstream mechanistic node rather than a biomarker.⁷³ Nicotine was also reported to promote cardiomyocyte apoptosis through oxidative stress and altered expression of apoptosis-related genes, supporting its involvement in cytotoxic cellular programming in cultured cardiomyocytes.⁹⁹ Nicotine can disrupt mitochondrial quality control by increasing Drp1-mediated fission and impairing mitophagic flux in the cardiomyocytes through ROS-dependent activation of stress kinases p38/Jun N-terminal kinase (p38/JNK) and reduced lysosomal protease activity.⁹³ It has been consistently reported in cardiac tissues that ROS and mitochondrial dysfunction feed forward into maladaptive metabolic reprogramming that includes impaired oxidative phosphorylation, altered mitochondrial dynamics, and reduced bioenergetic capacity.^{73,89,93} Notably, mechanistic directionality is context-dependent; in isolated mitochondrial preparations, nicotine was shown to interact with electron transport chain components and directly modulate superoxide generation.¹⁰⁰ Thus, existing studies support the concept that nicotine acts as a potent driver of redox imbalance across renal, vascular, and cardiac tissues, with mitochondria and NADPH oxidases emerging as dominant sources of pathological ROS. However, despite extensive associations with oxidative stress, the

precise molecular species, subcellular sources, and temporal dynamics of nicotine-induced ROS remain poorly defined, limiting mechanistic resolution.

NICOTINE, NOS SIGNALING, AND THE SHIFT FROM NO BIOACTIVITY TO NITROSATIVE STRESS

A consistent theme across vascular, cardiac, and, to a lesser extent, renal studies is that nicotine exposure is associated with reduced NO bioavailability and a pathological shift in nitric oxide synthase (NOS) output toward production of reactive nitrogen species, particularly peroxynitrite (ONOO⁻) (a process commonly termed nitrosative stress).^{92,101} Mice studies have shown that nicotine can impair NO signaling both by limiting NO production (eg, diminished eNOS function) and by accelerating NO consumption through reactions with superoxide (O₂⁻) to generate ONOO⁻, leaving tissue footprints such as 3-nitrotyrosine.^{101–103} A direct mechanism linking nicotine to endothelial oxidant stress is tetrahydrobiopterin (BH4) depletion-driven eNOS (NOS3) uncoupling, where eNOS shifts from generating NO to generating superoxide; nicotine reduces GTPCH1 (GTP cyclohydrolase-1, the rate-limiting enzyme for BH4 synthesis), increases ROS, and induces endothelial dysfunction, which is reversible by BH4/sepiapterin or GTPCH1 restoration.⁹² This mechanism is biologically consistent with broader NOS biochemistry showing that inadequate BH4 shifts eNOS from producing NO toward producing superoxide, thereby amplifying oxidative/nitrosative stress.^{104,105} In addition, nicotine has been linked to induction of inducible NOS (NOS2) in the vasculature, which can increase NO flux under inflammatory/stress conditions; in the presence of elevated superoxide (eg, during redox stress resulting from dysfunctional mitochondria), this would favor peroxynitrite formation and nitrative modification of proteins.^{106,107} In a murine arterial stiffness model, a SIRT1 gain of function protected against nicotine-induced arterial stiffening, and nicotine was shown to increase vascular inducible NO and 3-nitrotyrosine and was mechanistically connected to peroxynitrite-mediated SIRT1 inactivation, thereby placing reactive nitrogen species as a functional driver of disease-related vascular remodeling.¹⁰¹ Nicotine can alter NO biology beyond the vasculature, including modulation of NO metabolite levels and NOS-related signaling in the brain, indicating that nicotine-NOS interactions are not limited to one tissue and may be sex- and region-dependent.^{108,109} These broader observations reinforce that nicotine can intersect with NOS pathways at multiple regulatory levels (expression, coupling state, and substrate/cofactor availability), and that the overall phenotype likely depends heavily on the local redox environment. In renal injury contexts, nicotine exposure has been associated

with increased nitrotyrosine in kidney tissue, consistent with nitrative stress and peroxynitrite biology during ischemic or inflammatory injury. For example, chronic nicotine exposure exacerbated renal ischemia/reperfusion injury and increased kidney nitrotyrosine levels alongside other oxidative stress markers.¹⁷ While these data do not fully resolve which NOS isoform predominates in each renal compartment, they support a model in which nicotine increases the probability of a NO-to-ONOO⁻ shift in vivo, particularly under concurrent ROS-generating conditions that are common in CKD and acute injury. Moreover, nicotine is a source for Tobacco-Specific Nitrosamines, such as nicotine-derived nitrosamine ketone and N-nitrosornicotine.¹¹⁰ While both nicotine-derived nitrosamine ketone and N-nitrosornicotine are formed during tobacco product manufacturing,¹¹⁰ the transformation of nicotine into Tobacco-Specific Nitrosamines also occurs endogenously.^{111,112} Tobacco-Specific Nitrosamines have been linked to various cancers; moreover, their noncarcinogenic effects include effects on cellular metabolism, oxidative stress, and inflammation.¹¹³

KEY UNRESOLVED QUESTIONS AND GAPS IN KNOWLEDGE

In summary, existing studies strongly associate nicotine exposure with oxidative and nitrosative stress across renal, vascular, and cardiac tissues; however, the field remains focused on receptor-linked signaling, while direct, non-receptor-mediated actions of nicotine are rarely interrogated. This gap is particularly important for understanding chronic and long-lasting tissue damage, where sustained intracellular accumulation, mitochondrial interactions, and transporter-mediated handling of nicotine may be more relevant than acute nAChR activation. An important limitation in assessing the evidence of nicotine effects on oxidative and nitrosative stress is that the term ROS is used broadly and imprecisely in the literature, with limited resolution of species, subcellular sources, timing, and downstream targets, hindering causal inference. Finally, understanding nicotine's actions and evaluating the safety profile of NRPs are limited by the relative scarcity of studies examining nicotine alone. The majority of available evidence is derived from research on traditional combustible cigarettes or electronic cigarettes, where nicotine is delivered alongside numerous other bioactive compounds, making it difficult to attribute the observed effects specifically to nicotine. Importantly, a limited number of studies on the cardiovascular effects of NRPs are available. More focused, well-controlled studies using nicotine alone at physiologically and therapeutically relevant doses are needed to clearly define its independent mechanistic role in cardiovascular and renovascular pathology. Moreover, while the studies provide

evidence of nAChR-dependent action of nicotine, nicotine's receptor-independent action remains unexplored.

Another understudied factor in investigating nicotine-associated mechanisms is genetic variability in nicotine metabolism and the influence of genetics on susceptibility to nicotine-mediated injury. Variations in CYP2A6 (Cytochrome P450 2A6), primary enzyme responsible for nicotine metabolism contribute to differences in smoking cessation success. Common and rare CYP2A6 alleles (null, reduced- and increased-function, and copy-number variants) produce large variability in nicotine clearance and the nicotine metabolite ratio.^{114,115} Moreover, twin studies suggest nicotine dependence, withdrawal symptoms, and smoking behaviors are heritable up to 60%.^{116–118} However, no data are available regarding genetic variability in nicotine-mediated organ injury development. In addition, major nicotine transporters (OCT) expression is highly polymorphic, potentially influencing nicotine distribution across tissues. For instance, a meta-analysis showed an association of the OCT2 *Ser270Ala* polymorphism with success of smoking cessation therapy.¹¹⁹

Although there is a clear need to evaluate long-term mechanisms of smoking-associated injury, mechanistic data on postcessation states, as well as animal models of these states are scarce. It is well established clinically that former smokers retain elevated cardiovascular risk for years after quitting,³⁰ yet the molecular mechanisms and potential therapeutic targets are unclear. Whether this reflects persistent nitrosative stress, including ONOO[−]-mediated protein modifications and SIRT1 inactivation, sustained eNOS uncoupling driven by residual BH4 depletion; irreversible structural remodeling of the vascular wall; enduring oxidative enzyme upregulation such as Nox5 or inducible NO; or epigenetic reprogramming of redox-sensitive pathways has not been mechanistically characterized in postcessation states. While models of nicotine withdrawal exist,¹²⁰ long-term studies of associated tissue damage and cell-specific mechanisms in cardiovascular and renal systems are limited.

Addressing these limitations requires more mechanistically rigorous studies that move beyond generic redox stress toward quantitative, compartment-specific definitions of signaling driven by nicotine itself, independent of receptor activation, which will inform therapeutic approaches to nicotine-mediated cardiovascular and renal damage.

CONCLUSIONS AND FUTURE DIRECTIONS

Despite smoking being the leading preventable cause of death, responsible for 480 000 deaths annually in the United States,¹²¹ clinical and academic research has historically focused on cigarette and vape components

rather than nicotine itself. Nicotine consumption inflicts global damage, targeting almost every major organ of the body and resulting in at least 18 forms of cancer, and accelerating multiple cardiorenal and vascular pathologies, including atherosclerosis, myocardial infarction, coronary spasm, stroke, aortic aneurysm, peripheral obstructive arterial disease, arrhythmias, hypertensive heart disease, CKD, and heart failure.^{59,122} Nicotine is strongly linked to cardiovascular disease, hypertension, and CKD,^{16–18} yet its role in cardiovascular and kidney pathology remains poorly understood, and no targeted therapies exist for former smokers. Smoking cessation is the only recommended intervention, yet it has a low success rate and often leads to prolonged, continuous, and undocumented nicotine consumption. While quitting significantly reduces associated health risks, ≈80% of smokers attempting to quit on their own relapse within the first 30 days, and only about 3% remain abstinent at 6 months.¹²³ Furthermore, nicotine-induced chronic injury and prolonged inflammation impair tissue regeneration,¹²⁴ while in organs with limited regenerative capacity, such as the kidneys, nicotine-mediated damage is irreversible. Nicotine poses a significant risk in patients with CKD; however, no standardized smoking cessation guidelines exist for this population. Most nicotine cessation therapies contain nicotine, and only 2 FDA-approved options are nicotine-free. There is an urgent need for therapeutic strategies to mitigate persistent nicotine-induced vascular and kidney damage. Further mechanistic studies are essential for their development.

Collectively, the body of work reviewed here establishes nicotine as an active pathogenic driver of cardiovascular and renal injury rather than a benign addictive compound. However, the existing literature is weighted toward vascular mechanisms, with far fewer mechanistic studies interrogating renal parenchymal cells, particularly podocytes and proximal tubules, which are directly exposed to filtered nicotine and its metabolites and are central to CKD progression. Even within the cardiovascular field, most studies emphasize nAChR-dependent signaling, while direct, receptor-independent intracellular actions of nicotine (including membrane diffusion and transporter-mediated uptake remain poorly characterized.

Several critical gaps emerge that should shape future research directions. First, there is a striking absence of sex-specific data, despite well-established differences in nicotine metabolism, NO signaling, mitochondrial biology, and cardiorenal disease susceptibility between males and females. Mechanistic studies in the kidney should move beyond vascular effects to rigorously examine epithelial metabolic pathways, including how nicotine alters bioenergetics, redox buffering, and cell-to-cell crosstalk along the nephron. The pervasive use of the umbrella term oxidative stress in the literature has limited mechanistic resolution; future work must distinguish specific

reactive species (eg, superoxide, hydrogen peroxide, peroxynitrite), defined sources (mitochondria, NADPH oxidase isoforms, uncoupled NOS), and subcellular compartments, as well as temporal dynamics that differentiate adaptive from maladaptive signaling in response to nicotine. Importantly, emerging evidence implicates nitrosative stress and NOS uncoupling as functional drivers of lipid peroxidation, as underappreciated mechanisms of nicotine-induced injury that may be particularly relevant in chronic disease states such as hypertension, diabetes, and CKD. Dissecting how nicotine shifts NO signaling toward peroxynitrite generation, protein nitration, and irreversible metabolic remodeling (independent of receptor activation) represents a high-priority direction. Additional future efforts should include the development of postcessation models that reflect hypertension and CKD development, and incorporation of clinically relevant nicotine dosing paradigms, including NRPs and passive exposure. Integration of human genetics will be essential to understand interindividual susceptibility to nicotine-induced damage. Finally, identification of therapeutic targets that specifically mitigate nicotine-induced redox and nitrosative tissue injury needs to be at the center of attention. Addressing these gaps is essential for moving the field from associative observations to actionable mechanisms and for developing strategies to protect vulnerable cardiorenal populations who remain at risk long after smoking cessation.

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